CTF WRITEUP

Havoc C2 Framework — Network Forensics

TryHackMe — Packet Analysis / Wireshark – Achira Thilakarathne

1. Overview

Havoc is a modern, open-source, and malleable post-exploitation command and control (C2) framework designed for red teaming and penetration testing and often used as a actual tool for exploiting and malicious activity.

The hex value deadbeef is present in the captured packets, identifying the Command & Control framework as Havoc C2

How to locate the marker in Wireshark:

Edit → Find Packet → Search by Hex Value → enter: deadbeef

2. Packet 182 — Initial Beacon (Client → Server)

Packet 182 contains the first occurrence of the deadbeef magic bytes. This is a POST request from 10.0.2.38 to 10.0.2.37, confirming 10.0.2.38 is the client beaconing to the C2 server.

Action: Copy Hex Stream from Packet 182

Hex Stream Breakdown

```
00000113
deadbeef
0e9f (skip)
b7d8
0000006300000000 (skip)
946cf2f65ac2d2b868328a18dedcc296cc40fa28fab41a0c34dcc010984410ca  ← AES Key
8cd00c3e349290565aaa5a8c3aacd430  ← IV Key
```

Data field (encrypted beacon payload — excerpt):

```
340cd2bc567802a30e1c2722bb2135dbfe9b76cc0f486e975aaf1f21a6ad13f9
ef282131470a0191c974b4f5c2369005227811dcfb9708192e358fe9dfbf4214
760291da15237a0f2e5db5b1e718d48ef01af5e724767c95a1b5c136cc744d75
edcb1747847d1b6a22df5a23531442b73329c2b2ae46e1ab2506f548bd1b7068
... (full stream in Wireshark)
```

HTTP headers (hex-decoded from same packet):

```
POST / HTTP/1.1
Cache-Control: no-cache
Connection: Keep-Alive
Pragma: no-cache
Content-Type: */*
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/96.0.4664.110 Safari/537.36
Content-Length: 5781
Host: 10.0.2.37
```

3. Packet 239 — SID Extraction

The hex stream from Packet 239 was decrypted using the AES key and IV extracted from Packet 182. The decrypted output reveals the victim's Windows Security Identifier (SID).

Note: For client→server packets, copy Hex Stream. For server→client packets, copy the Data field.

Hex stream (excerpt — full stream in Wireshark):

```
0e9fb7d8000000001000000003a93653eecbbd285595568438269
0ca5bea610e8236938e0298f5301e78d13978941465456071c9d
c86eca9abd4af87d497811d489f96d643f64c5b881f909556e47
e5b2361214505a3bb8eeea61f4cbd02b9afa4a4f1f91c8bbaf78
... (truncated)
```

Extracted SID:

```
S-1-5-21-679395392-3966376528-1349639417-1103
```

4. Server Network Information

The hex stream from the server reply (copy Data field, not Hex Stream) was decrypted to reveal the server's full network configuration.

Field	Value
Adapter	`Ethernet`
DNS Suffix	`home`
Link-local IPv6	`fe80::e134:1b0c:c8d5:3020%6`
IPv4 Address	`10.0.2.38`
Subnet Mask	`255.255.255.0`
Default Gateway	`10.0.2.1`

5. Packet 314 — System Information

Decrypted output from Packet 314 reveals the full systeminfo output of the compromised host.

Field	Value
Host Name	`WIN-9H86M71MBE9`
OS Name	`Microsoft Windows Server 2019 Standard Evaluation`
OS Version	`10.0.17763 N/A Build 17763`
OS Configuration	`Primary Domain Controller`
System Manufacturer	`innotek GmbH`
System Model	`VirtualBox`
System Type	`x64-based PC`

Field	Value
Processor	`Intel64 Family 6 Model 158 Stepping 13 GenuineIntel ~3600 MHz`
Windows Directory	`C:\Windows`
System Locale	`en-us; English (United States)`
Time Zone	`(UTC-08:00) Pacific Time (US & Canada)`
Total Physical Memory	`8,192 MB`
Domain	`clientserver.thm`
Logon Server	`\\WIN-9H86M71MBE9`
Hotfixes Installed	`KB5020627, KB5019966, KB5020374`
NIC	`Intel(R) PRO/1000 MT Desktop Adapter`
DHCP Server	`10.0.2.3`

6. Flags & Credentials Found

Flag 1 — Packet 341 (Printed Output)

```
THM{HavOc_C2_DeCRypTing_is_Fun_Fun_FUN}
```

Flag 2 — Packet 610 (File Exfiltration)

Important file path (Packet 602):

```
C:\Users\paco\Desktop\Files\clients.csv
```

```
THM{I_Can_SEE_ThE_fiL3_YoU_ToOk}
```

Attacker-Created Credentials

Username	Password
administrato	`WfD3hz3AXZ4n`